MILITARY SPECIFICATION

FORGING, MOTOR BODY M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for motor body forgings used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-6868	Inspection Process, Magnetic Particle
MIL-H-6875	Heat Treatment of Steel (Aircraft Practice) Process for
MIL-F-7190	Forging, Steel, Carbon and Alloy (For Aircraft Application)
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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STANDARDS

FEDERAL

FED-STD-151	Metals: Test Methods

MILITARY

MIL-STD-410	Non-Destructive Testing Personnel Qualification and Certification
MIL-STD-1188	Commercial Packaging of Supplies and Materials

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9328316	Forging, Motor Body, 8 Inch, M753
9328453	Forging, Motor Body, 8 Inch (Test Sample)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8	Tension Testing of Metallic Materials

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Construction. Forgings shall comply with all requirements specified on drawing (dwg.) 9328316, with all applicable requirements of MIL-F-7190 (Press Die-Grade A), with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the dwg. and herein. Forgings shall be delivered in the annealed state as required by the dwg. but capable of meeting the physical properties herein after hardening.

3.2 Material. Forging material shall comply with all requirements specified on dwg. 9328316. Testing shall be performed in accordance with 4.5.1. The contractor shall provide objective evidence (see 6.4) that forging material complies with all dwgs. and specification requirements.

3.3 Physical defects. Forgings shall exhibit no evidence of cracks, flaws, unsound areas or other obvious physical defects when subjected to magnetic particle inspection. Surface defects shall not exceed .010 inches in any dimension. Testing shall be performed in accordance with 4.5.2.

3.4 Physical properties.

3.4.1 Yield strength and elongation. When heat treated, quenched and tempered in accordance with dwg. 9328453 and the applicable requirements of MIL-H-6875, forgings shall be capable of attaining a minimum yield strength of 180,000 pounds per square inch (psi) at 0.2% offset and a minimum elongation of 10%. Testing shall be performed in accordance with 4.5.3.

3.4.2 Grain flow. When heat treated, quenched and tempered in accordance with dwg. 9328453 and the applicable requirements of MIL-H-6875 the grain flow of sectioned forgings shall lie in the same general direction as the longitudinal axis of the forging except in corners and other high stress areas where the grain shall flow in a manner which will minimize the stress in these areas (see dwg. 9328316 for typical grain flow). Testing shall be performed in accordance with 4.5.4.

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3.5 First article inspection. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.6 Workmanship. Forgings shall be free of tears, seams, laps, embedded scale, hard spots, overheated areas or any other obvious physical defects likely to adversely affect the items intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

1. First Article Inspection (see 4.3)
2. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.5), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Forging, Motor Body, 8 Inch, M753	9328316	5

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All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be obtained from the same source of supply as will be used in regular production. Test equipment, as required, shall be approved by the procuring activity prior to submission of the first article sample.

4.3.2 Inspections to be performed. See MIL-A-48078 and the first article inspection table herein.

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63173(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.3.2.1	Forging, Motor Body, 8 Inch, M753		1 OF 1		9328316 & 9328453
					NEXT HIGHER ASSEMBLY: M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classification of defects (see note 1).	5			4.4.2.1
	Dimensional characteristics & workmanship Material Physical properties (magnetic particle inspection) Yield strength and elongation (see notes 2 and 3) Grain flow (see note 3)				

NOTE 1. Tests shall be performed in the order indicated.

NOTE 2. The 5 items for yield strength and elongation tests shall be inspected for dimensional characteristics of dwg. 9328453 prior to performing yield strength and elongation tests.

NOTE 3. Yield strength, elongation and grain flow are destructive and shall be performed last by sectioning 5 units approximately in half and performing grain flow examination on one half and yield strength and elongation tests on the other half of the sections.

NOTES:

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4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, an inspection lot shall consist of forgings produced from one heat of material in one unchanged process with the same set of forging dies. Any repair to or change in dies shall necessitate formation of a new inspection lot.

4.4.2 Examinations. Examination shall be in accordance with MIL-A-48078 and the quality conformance inspection tables herein. Unless otherwise specified in the quality conformance inspection tables herein, sampling plans for Major and Minor defects shall be in accordance with Tables I and II, as applicable.

Table I. Sampling plan for major defects.

LQ = .05 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table II. Sampling plan for minor defects.

LQ = .10 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63173(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Forging, Motor Body, 8 Inch, M753	1 OF 4	9328316 & 9328453
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Overall outside diameter		Table I	3.1	Gage
102.	Runout of outside diameter		Table I	3.1	Gage
103.	Inside diameter (6.61)		Table I	3.1	Gage
104.	Roundness of inside diameter		Table I	3.1	Gage
105.	Length bulkhead to end (9.19 min.)		Table I	3.1	Gage
106.	Thickness of bulkhead		Table I	3.1	Gage
107.	Perpendicularity of bulkhead		Table I	3.1	Gage
108.	Outside diameter of delay area (3.02)		Table I	3.1	Gage
109.	Runout of outside diameter of delay area		Table I	3.1	Gage
110.	Taper inside of boattail section, 2 places (0° + 3°)		Table I	3.1	Gage
111.	Outside diameter of boattail section (7.65)		Table I	3.1	Gage
112.	Runout of boattail section		Table I	3.1	Gage
113.	Radius inside of boattail section (1.12)		Table I	3.1	Gage
114.	Radius inside of boattail section (.75)		Table I	3.1	Gage

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63173 (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.1 (cont)	Forging, Motor Body, 8 Inch, M753		2 OF 4	9328316 & 9328453
				NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major: (cont)					
115.	Radius outside of boattail section (.38 max)		Table I	3.1	Gage
116.	Inside diameter of boattail section (6.26)		Table I	3.1	Gage
117.	Runout of inside diameter of boattail section		Table I	3.1	Gage
118.	Taper of outside diameter of boattail section (5°)		Table I	3.1	Gage
119.	Depth of boattail section (2.34)		Table I	3.1	Gage
120.	Depth of delay area (1.81)		Table I	3.1	Gage
121.	Radius in delay area (.12)		Table I	3.1	Gage
122.	Corner outside boattail section, optional corner only (.25)		Table I	3.1	Gage
123.	Parallelism of rear of boattail section to bottom of cavity		Table I	3.1	Gage
124.	Parallelism of bottom of boattail section to bottom of cavity		Table I	3.1	Gage
125.	Radius inside of cavity (.75)		Table I	3.1	Gage

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PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER
4.4.2.1 (cont)	Forging, Motor Body, 8 Inch, M753				9328316 & 9328453
					NEXT HIGHER ASSEMBLY
					M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major: (cont)					
126.	Radius outside boattail section (.50 max)		Table I	3.1	Gage
127.	Physical properties (magnetic particle inspection)		100%	3.3	4.5.2
128.	Yield strength and elongation (see notes 1 and 2)	5/ lot		3.4.1	4.5.3
129.	Grain flow (see note 2)	5/ lot		3.4.2	4.5.4
130.	Material	5/ lot		3.2	4.5.1
Minor:					
201.	Evidence of poor workmanship		Table II	3.6	Visual
202.	Surface finish 500		Table II	3.1	Visual
203.	Marking missing or incorrect		Table II	3.1	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63173(AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.1 (cont)	Forging, Motor Body, 8 Inch M753		4 OF 4	9328316 & 9328453
				NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	NOTE 1. The 5 units/lot shall be inspected for the dimensional characteristics of dwg. 9328453 prior to performing yield strength and elongation tests. NOTE 2: Yield strength, elongation and grain flow are destructive and shall be performed by sectioning 5 units/lot approximately in half and performing grain flow examination on one half and yield strength and elongation tests on the other half of the sections.				

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see MIL-A-48078 and 6.3 herein).

4.5 Test methods.

4.5.1 Material (see 3.2). Each heat of forging material shall be analyzed in accordance with FED-STD-151, Method 111.2 or Method 112.2.

4.5.2 Physical defects (see 3.3). Each forging shall be examined for physical defects in accordance with MIL-I-6868 or an approved equivalent using the continuous wet fluorescent method. Only circular magnetization induced by a central conductor will be required. The suspension shall be applied to all surfaces of the forging. The magnetic solid content of the suspension shall be maintained at 0.2 to 0.4 mililiters (ml) of solids per 100 ml of suspension when oil is used and 0.15 to 0.35 ml of solids per 100 ml of suspension when water is used. Three applications of the magnetic field shall be made on the forging with a minimum of 2500 amperes per application. The forging shall be rotated 120 degrees after each application. After draining, the forging shall be inspected under ultra-violet light for the presence of defects. The requirement of MIL-I-6868 to check system effectiveness by means of a steel ring specimen shall not apply but system effectiveness shall be checked against approved standards at the start of each days production and at eight hour intervals. If the inspection process is determined to be inadequate the inadequate condition shall be corrected and all forgings inspected during the previous eight hour interval shall be reinspected. Inspection shall be performed by operators certified in accordance with MIL-STD-410.

4.5.3 Yield strength and elongation (see 3.4.1). After heat treating, quenching and tempering, in accordance with dwg. 9328453, the forging yield strength and elongation shall be tested at the locations specified on dwg. 9328453 in accordance with ASTM E8 or equivalent. Testing shall be performed on the largest substandard size specimen possible.

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4.5.4 Grain flow (see 3.4.2). After heat treating, quenching and tempering, the forging shall be sectioned (approximately in half) with the section cut parallel to the longitudinal axis and the exposed surface machined and cleaned as required to obtain a smooth, polished surface. The section shall then be placed in a warm acid bath of hydrochloric acid and water (approximately half and half) for five or ten minutes or otherwise etched by an approved method. The section shall then be cleaned and examined for the grain flow direction.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified in the contract, packaging, packing and marking shall be in accordance with MIL-STD-1188 (commercial pack).

6. NOTES

6.1 Intended use. The forgings covered by this specification are intended for fabrication of rocket motor bodies used on M753, 8 Inch Rocket assisted projectile.

6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and data of this specification.
b. First article inspection, when required (see 3.5).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

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(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Objective evidence. A statement (certification), supported by test data, that all materials produced or purchased by the contractor meets requirements when such material is controlled by Government or commercial specifications referenced in contractual documents.

6.5 Supersession data. This specification includes the requirements of AR-PD-14 dated 5 February 1979.

6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading US Army Armament Research and Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal, dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:

Army-AR

Preparing activity:

Army-AR

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